

Test Run Report — Run #123

Environment: Development • Trigger: Scheduled • Started: Aug 10, 2026 01:41 • Ended: Aug 10, 2026 04:24
Generated: 8/10/2026, 4:28:31AM

Summary

231

Total Tests

92.6%

Pass Rate (excl. skipped)

175 / 14

Passed / Failed

42

Skipped (connection)

Failures (14)

Failure Cluster	Category	Failures
LLM Reasoning Issues	Llm Reasoning	9
Data Availability Issues	Data Availability	2
Tool Execution Failures	Tool Execution	2
Graph/Visualization Errors	Graph Visualization	1

LLM Reasoning Issues LLM REASONING (9 failures)

Question: Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt

Error reason: AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 05:42 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

TECHNICAL EVIDENCE

Exception	AttributeError: 'NoneType' object has no attribute 'rule_code'
Agent/Tool	data_agent / extract_timeseries_data_ag
Endpoint	/testuser1/ai_agent_chat
Location	agent_graph.py:916

API call: GET /testuser2/ai_agent_chat → 400

Raw API response:

```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified
and will investigate.",
  "coding_response": [
    "===== ERROR SUMMARY =====\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGEN
T: data_agent\nTOOL: extract_timeseries_data_ag\nERROR TYPE: AttributeError\nERROR MESSAGE: 'NoneType' obje
ct has no attribute 'rule_code'\nLOCATION: agent_graph.py:916\n=====
  ]
}
```

```
===== ERROR SUMMARY =====
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: extract_timeseries_data_ag
ERROR TYPE: AttributeError
ERROR MESSAGE: 'NoneType' object has no attribute 'rule_code'
LOCATION: agent_graph.py:916
=====
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)



Are you satisfied with the answer?



Time taken: 17s, token: 0 tokens

Please find out the daily POA poa summed up as a Wh/m^2 , daily for the last 12 months and then calculate the median, mean and kurt



Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.

05:42 AM



Do you want to report this error?

YES NO

Time taken: s, token: tokens

● connected to Mathkor Gid



How can we help you today?



LEAVE ANY
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Error reason: AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 05:43 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

TECHNICAL EVIDENCE

Exception	PydanticSerializationError: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>
Agent/Tool	data_agent / unknown
Endpoint	/testuser1/ai_agent_chat
Location	agent_graph.py:916

API call: GET /testuser2/ai_agent_chat → 400

Raw API response:

```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.",
  "coding_response": [
    "===== ERROR SUMMARY =====\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGENT: data_agent\nTOOL: unknown\nERROR TYPE: PydanticSerializationError\nERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>\nLOCATION: agent_graph.py:916\n=====
  ]
}
```

```
===== ERROR SUMMARY =====
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: unknown
ERROR TYPE: PydanticSerializationError
ERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>
LOCATION: agent_graph.py:916
=====
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)



Do you want to report this error?

YES NO

Time taken: s, token: tokens

Is there shading happening with my plant?
Please analyze the pyranometer and check for frequent dips at certain times



Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.

05:43 AM



Do you want to report this error?

YES NO

Time taken: s, token: tokens

● connected to Mathkor Gid



How can we help you today?



LEAVE ANY
FEEDBACK



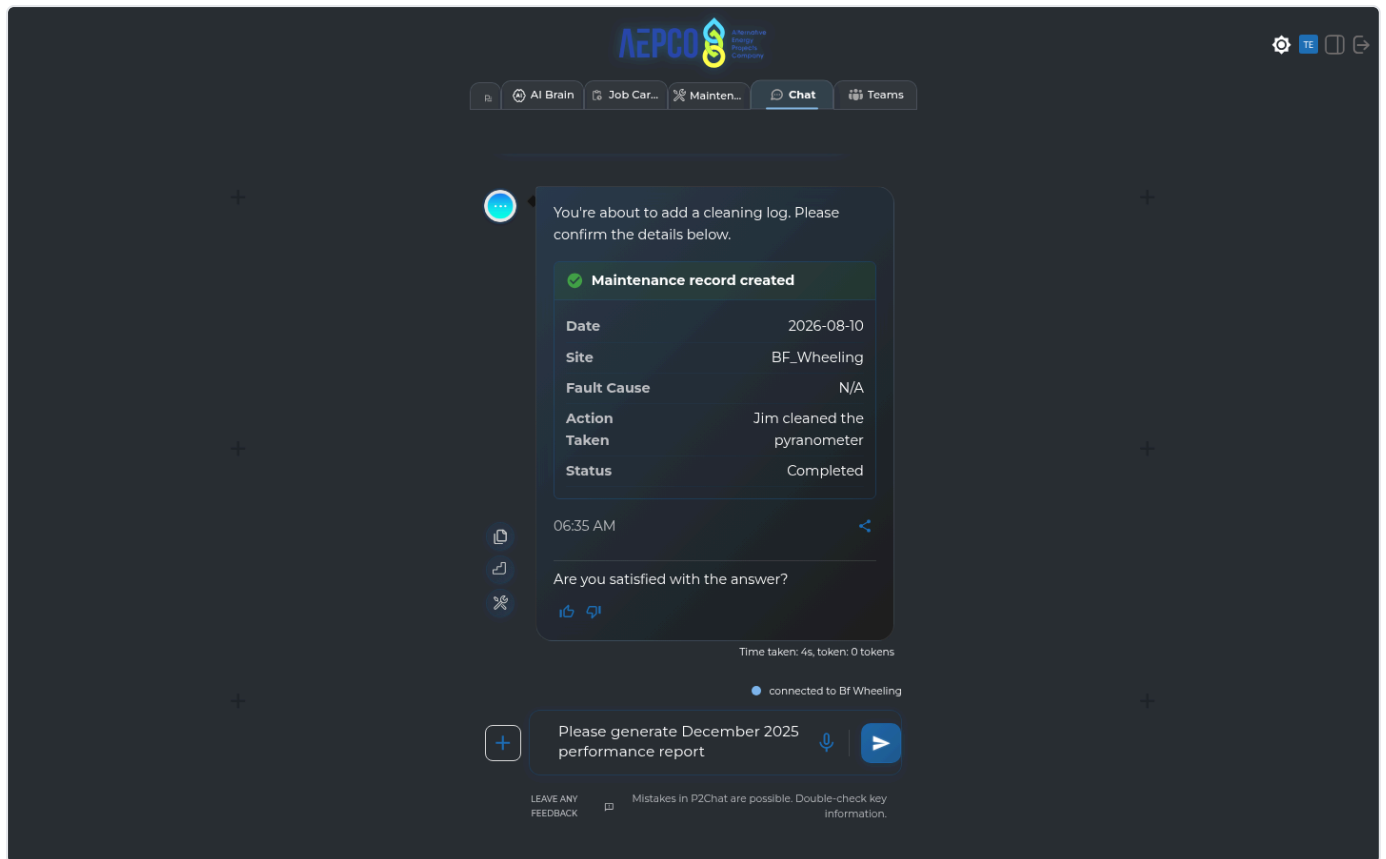
Mistakes in P2Chat are possible. Double-check key information.

3. BF_Wheeling

Development • 17.0s

Question: Please generate December 2025 performance report

Error reason: AI indicated failure (matched 'unable to provide'): AI Brain Job Cards Maintenance Chat Teams Active Power is 0W Wind Speed is 0m/s Weather Temperature is 17°C Last updated: 8/10/2026, 03:00 AM 06:26 AM Good evening. How can I help you tod



4. Abdali_Mall_Netmetering

Development • 21.5s

Question: When was the last time the BOM data showed up

Error reason: AI indicated failure (matched 'i'm sorry'): I'm sorry, but I am only able to assist with user interface actions, settings, and guides. For specific data reports like BOM data occurrences, please use the report module or ask the plant performance

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "When was the last time the BOM data showed up",
    "displayed_response": "I'm sorry, but I am only able to assist with user interface actions, settings, and guides. For specific data reports like BOM data occurrences, please use the report module or ask the plant performance report agent.",
    "coding_response": [
      [
        {
          "answer": "I'm sorry, but I am only able to assist with user interface actions, settings, and guides. For specific data reports like BOM data occurrences, please use the report module or ask the plant performance report agent.",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "ui_agent"
    ],
    "used_tools": [],
    "urls": [],
    "steps": []
  }
}
```



Are you satisfied with the answer?



Time taken: 5s, token: 0 tokens

When was the last time the BOM data showed up



I'm sorry, but I am only able to assist with user interface actions, settings, and guides. For specific data reports like BOM data occurrences, please use the report module or ask the plant performance report agent.

06:51 AM



Are you satisfied with the answer?



Time taken: 6s, token: 0 tokens

connected to Abdali Mall Netmetering



How can we help you today?



LEAVE ANY
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Error reason: AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 06:59 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

TECHNICAL EVIDENCE

Exception	PydanticSerializationError: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>
Agent/Tool	data_agent / unknown
Endpoint	/testuser1/ai_agent_chat
Location	agent_graph.py:916

API call: GET /testuser2/ai_agent_chat → 400

Raw API response:

```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.",
  "coding_response": [
    "===== ERROR SUMMARY =====\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGENT: data_agent\nTOOL: unknown\nERROR TYPE: PydanticSerializationError\nERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>\nLOCATION: agent_graph.py:916\n=====
  ]
}
```

```
===== ERROR SUMMARY =====
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: unknown
ERROR TYPE: PydanticSerializationError
ERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>
LOCATION: agent_graph.py:916
=====
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)



Are you satisfied with the answer?



Time taken: 20s, token: 0 tokens

Is there shading happening with my plant?
Please analyze the pyranometer and check for frequent dips at certain times



Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.

06:59 AM



Do you want to report this error?

YES NO

Time taken: s, token: tokens

connected to Abdali Mall Netmetering



How can we help you today?



LEAVE ANY
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

6. Abdali_Mall_Lahoun

Development • 56.5s

Question: When was the last time the BOM data showed up

Error reason: AI indicated failure (matched 'an error'): Sorry, there was an error processing your request.
07:11 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected to Abdali Mall Lahoun LEAVE ANY FEEDBACK Mistakes in

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 500

Raw API response:

```
{
  "error": "An unexpected error occurred. Please try again later."
}
```

Console errors: Failed to load resource: the server responded with a status of 500 (Internal Server Error)

The screenshot displays the AEP CO chat interface. At the top, there is a navigation bar with the AEP CO logo and several tabs: 'AI Brain', 'Job Car...', 'Mainten...', 'Chat', and 'Teams'. The 'Chat' tab is currently selected. Below the navigation bar, the chat history is visible. It starts with a blue bubble containing the question 'When was the last time the BOM data showed up'. This is followed by a white bubble containing an error message: 'Sorry, there was an error processing your request.' Below the error message, there is a timestamp '07:11 AM' and a follow-up question 'Do you want to report this error?' with two options: 'YES' and 'NO'. At the bottom of the chat area, there is a text input field with the placeholder 'How can we help you today?' and a microphone icon. The interface also includes a 'TE' (Technical Evidence) button and a 'connected to Abdali Mall Lahoun' status indicator.

7. Abdali_Mall_Lahoun

Development • 17.0s

Question: Please generate December 2025 performance report

Error reason: AI indicated failure (matched 'an error'): AI Brain Job Cards 16 Maintenance Chat Teams
Active Power is 0W Wind Speed is 3.03m/s Weather Temperature is 27.29°C Last updated: 8/9/2026,
08:30 PM 07:04 AM Good evening. How can I hel

The screenshot displays the AEPCCO AI Brain interface. At the top, the AEPCCO logo is visible, along with navigation tabs for 'AI Brain', 'Job Car...', 'Mainten...', 'Chat', and 'Teams'. The 'Chat' tab is currently selected. A central modal window shows a confirmation message: 'You're about to add a cleaning activity log. Please confirm the details below.' Below this, a green checkmark indicates 'Maintenance record created'. A table lists the details of the record:

Date	2026-08-10
Site	Abdali_Mall_Lahoun
Fault	Pyranometer required
Cause	cleaning
Action Taken	Cleaned the pyranometer
Status	Completed
Responsibility	Jim

Below the table, the time '07:15 AM' is shown. A question 'Are you satisfied with the answer?' is followed by thumbs up and thumbs down icons. At the bottom of the modal, a status bar indicates 'Time taken: 4s, token: 0 tokens connected to Abdali Mall Lahoun'. Below the modal, a text input field contains the query 'Please generate December 2025 performance report' with a microphone icon and a send button. At the very bottom, a footer note reads: 'Mistakes in P2Chat are possible. Double-check key information.'

Question: How did my plant perform yesterday?

Error reason: AI indicated failure (matched 'an error'): For yesterday (2026-08-09), the actual measured plant performance metrics are: - Total energy produced: 5,394.91 kWh - Performance ratio (PR): 0.7834 - Specific yield: 6.8638 kWh/kWp - Refere

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "For yesterday (2026-08-09), the actual measured plant performance metrics are:\n\n- Total energy produced: 5,394.91 kWh\n- Performance ratio (PR): 0.7834\n- Specific yield: 6.8638 kWh/kWp\n- Reference yield (Yr): 7.1554 kWh/kWp\n- Expected yield (assuming PR target 0.8): 5.7243 kWh/kWp\n\nAvailability:\n- Technical/contractual availability could not be computed because the availability function failed.\n- Energy-based availability also could not be computed from the tool because it returned an error for this one-day query.\n\nData note:\n- The underlying data for 2026-08-09 only extends to 20:30:00, not the full day through 23:55:00. So these values are based on the available measured data up to 20:30.\n\nOther relevant indicator:\n- PR is slightly below the 0.80 expected benchmark implied by the expected-yield calculation.",
    "coding_response": [
      "For yesterday (2026-08-09), the actual measured plant performance metrics are:\n\n- Total energy produced: 5,394.91 kWh\n- Performance ratio (PR): 0.7834\n- Specific yield: 6.8638 kWh/kWp\n- Reference yield (Yr): 7.1554 kWh/kWp\n- Expected yield (assuming PR target 0.8): 5.7243 kWh/kWp\n\nAvailability:\n- Technical/contractual availability could not be computed because the availability function failed.\n- Energy-based availability also could not be computed from the tool because it returned an error for this one-day query.\n\nData note:\n- The underlying data for 2026-08-09 only extends to 20:30:00, not the full day through 23:55:00. So these values are based on the available measured data up to 20:30.\n\nOther relevant indicator:\n- PR is slightly below the 0.80 expected benchmark implied by the expected-yield calculation."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "multi_tool_use.parallel",
      "functions.final_result"
    ],
    "urls": [],
    "steps": "1. I choose data from August 9, 2026, including that day.\n2. I find all inverter power columns and their DC rated powers using the plant's metadata.\n3. I set calculation parameters like the irradiance threshold (usually 50 W/m²), power threshold (usually 0.1 kW), and data interval (typically 5 minutes).\n4. For each inverter, I filter the data to times when irradiance is above the threshold, then count the total useful time (T_useful) and downtime (T_down).\n5. I convert these counts into hours based on the data interval.\n6. I calculate Technical Availability for each inverter using: (T_useful - T_down) / T_useful × 100.\n7. I provide a dictionary showing the availability percentage for each inverter.\n\n---\n\n1. I select data from August 9, 2026, including that day.\n2. I identify the relevant columns for power, irradiance, and temperature.\n3. I clean the data by removing or flagging invalid power and irradiance values.\n4. I calculate the actual energy produced by multiplying power by the time interval (0.0833 hours) and adjusting for units.\n5. I calculate the expected energy without temperature correction.\n6. If the system is bifacial, I apply extra corrections based on metadata like ground albedo, area normalization, and shading losses.\n7. I group actual and expected energy by day.\n8. I calculate the performance ratio for each interval, applying bifacial corrections if needed.\n9. I return the average performance ratio across all intervals.\n10. I cap performance ratio values at 1.3 to avoid unrealistic results.\n\n---\n\n1. I select data from August 9, 2026, including that day.\n2. I calculate the total measured energy produced by the plant during this period.\n3. I identify and sum all major energy losses during this time, excluding baseline and measured energy, using a loss classification method.\n4. I calculate energy-based availability as the ratio of delivered energy to the sum of delivered and lost energy, shown as a percentage.\n5. I return the energy-based availability"
```

computed from the tool because it returned an error for this one-day query.

Data note:

- The underlying data for 2026-08-09 only extends to 20:30:00, not the full day through 23:55:00. So these values are based on the available measured data up to 20:30.

Other relevant indicator:

- PR is slightly below the 0.80 expected benchmark implied by the expected-yield calculation.

07:17 AM



Are you satisfied with the answer?



Time taken: 28s, token: 0 tokens

● connected to Abdali Mall Lahoun



How can we help you today?



LEAVE ANY
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Error reason: AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 07:19 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

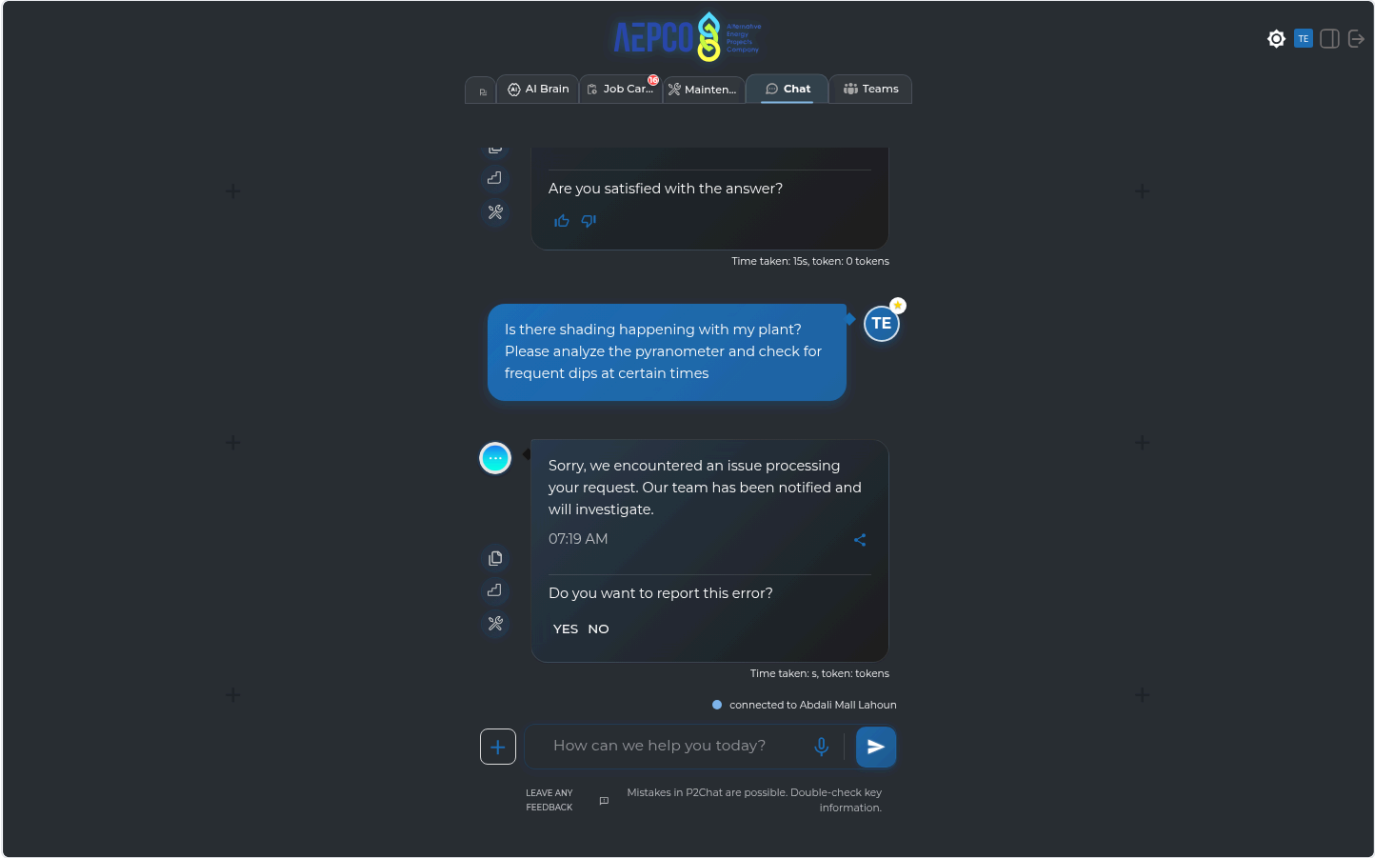
TECHNICAL EVIDENCE

Exception	PydanticSerializationError: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>
Agent/Tool	data_agent / unknown
Endpoint	/testuser1/ai_agent_chat
Location	agent_graph.py:916
API call:	GET /testuser2/ai_agent_chat → 400
Raw API response:	

```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.",
  "coding_response": [
    "===== ERROR SUMMARY =====\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGENT: data_agent\nTOOL: unknown\nERROR TYPE: PydanticSerializationError\nERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>\nLOCATION: agent_graph.py:916\n====="
  ]
}
```

```
===== ERROR SUMMARY =====
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: unknown
ERROR TYPE: PydanticSerializationError
ERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>
LOCATION: agent_graph.py:916
=====
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)



Data Availability Issues DATA AVAILABILITY (2 failures)

1. BF_Wheeling

Development • 54.9s

Question: Please generate last months report

Error reason: AI indicated failure (matched 'error occurred'): Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please generate last months report",
    "displayed_response": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar'",
    "coding_response": [
      [
        {
          "answer": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar'",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "report_agent"
    ],
    "used_tools": [
      "agent_external_report_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```



Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AL_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AI_Mushagar'

06:25 AM



Are you satisfied with the answer?



< Click here for Source

< Click here for Steps

< Click here for Tools

Time taken: 40s, token: 0 tokens

● connected to BF Wheeling



How can we help you today?



LEAVE ANY
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

2. BF_Wheeling

Development • 49.0s

Question: How did my plant perform yesterday?

Error reason: AI indicated failure (matched 'error occurred'): Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv.

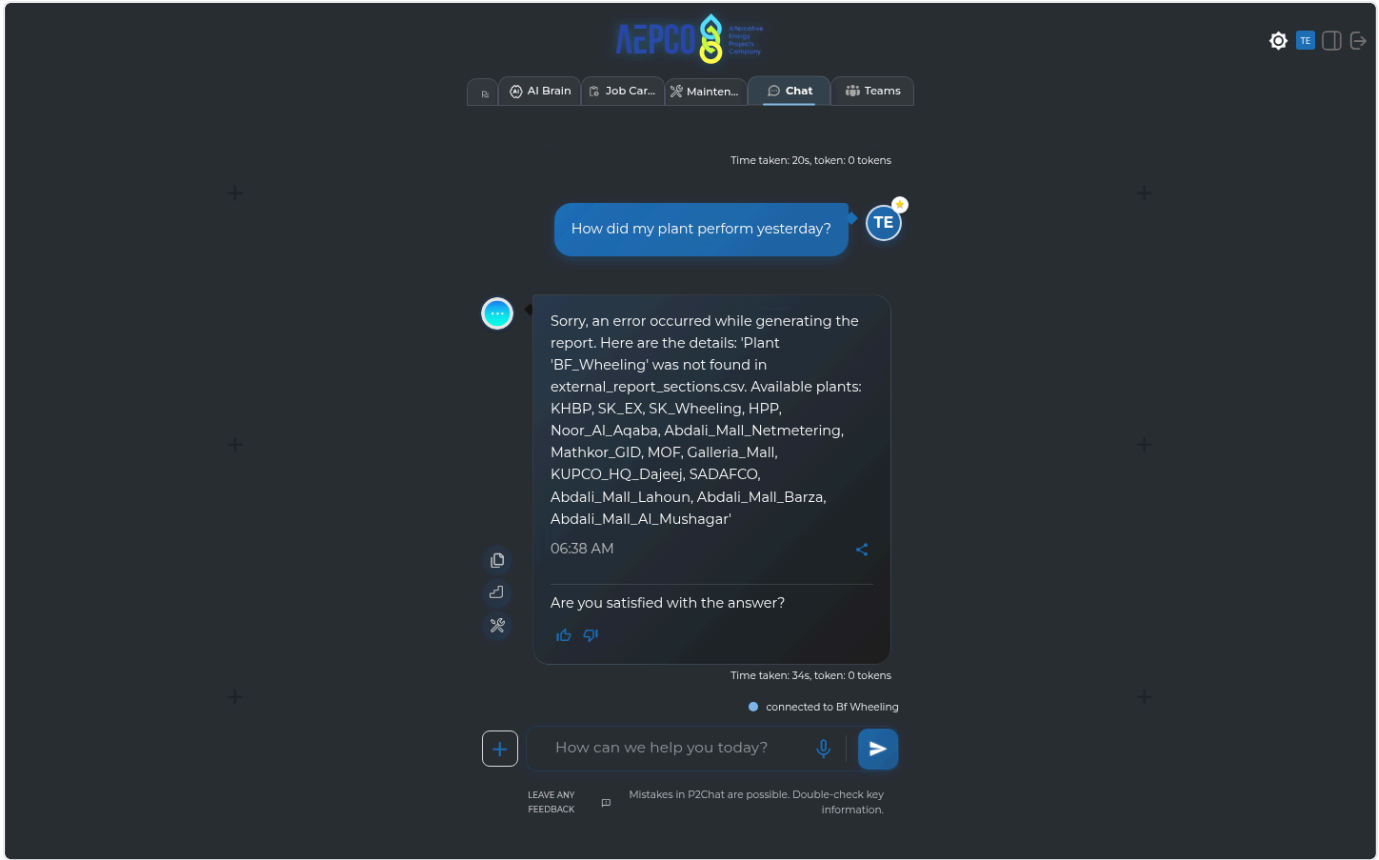
Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar'",
    "coding_response": [
      [
        {
          "answer": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_AI_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_AL_Mushagar'",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "report_agent"
    ],
    "used_tools": [
      "agent_external_report_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```



Tool Execution Failures TOOL EXECUTION (2 failures)

1. BF_Wheeling

Development • 44.5s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Error reason: AI indicated failure (matched 'internal error'): Clipping Loss Results: - No per-inverter clipping-loss values are available for the last 2 months due to a tool failure. Data Quality Notes: - The clipping-loss tool encountered an internal error:

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well",
    "displayed_response": "Clipping Loss Results:\n\n- No per-inverter clipping-loss values are available for the last 2 months due to a tool failure.\n\nData Quality Notes:\n\n- The clipping-loss tool encountered an internal error: unsupported operand type(s) for /: 'NoneType' and 'int' while building the digital twin.\n\n- As a result, the calculated values could not be generated and are not reliable.\n\nVisualization:\n\n- The chart \"Clipping losses for each inverter - last 2 months\" has been prepared.",
    "coding_response": [
      "I couldn't calculate the clipping losses because the clipping-loss tool failed internally for this date range with: `unsupported operand type(s) for /: 'NoneType' and 'int'` while building the digital twin.\n\nSo I can't provide trustworthy per-inverter clipping-loss values for the last 2 months from the available tools right now.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping losses for each inverter - last 2 months",
        "yAxisTitle": "Clipping Losses (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.calculate_clipping_or_curtailement_losses_ag"
    ],
    "urls": [],
    "steps": "1. Select data from June 10, 2026 to August 10, 2026, including the end date.\n2. Find all valid inverter power columns in the dataset, ignoring any columns related to reactive power or flags.\n3. Make sure the chosen power columns actually exist in the dataset.\n4. Clean the plane of array (POA) irradiance data by filling in missing values using a method that accounts for daily sunlight patterns.\n5. For each inverter power column, calculate power losses by finding the absolute difference between the measured power and the inverter's rated AC power during periods of detected loss.\n   - Detect clipping events using both a machine learning approach (rdtools XGBoost clip filter) and a simple threshold, flagging times when power is above 85% of the rated AC power.\n6. Convert the detected power losses into energy lost (in kWh) by multiplying by the time interval (0.0833 hours).\n7. Group the energy loss values by the chosen time period (such as day, month, or year) for each inverter.\n8. Add up the grouped values to get a total loss for each inverter over the selected period.\n9. Organize the results into a DataFrame, with inverters as columns and time periods as rows.\n10. Return the results as a dictionary, either broken down by time period or as a single total, depending on the settings."
  }
}
```

- No per-inverter clipping-loss values are available for the last 2 months due to a tool failure.

Data Quality Notes:

- The clipping-loss tool encountered an internal error: unsupported operand type(s) for /: 'NoneType' and 'int' while building the digital twin.
- As a result, the calculated values could not be generated and are not reliable.

Visualization:

- The chart "Clipping losses for each inverter – last 2 months" has been prepared.

06:40 AM



Are you satisfied with the answer?



Time taken: 29s, token: 0 tokens

● connected to BF Wheeling



How can we help you today?



LEAVE ANY
FEEDBACK



Mistakes in P2Chat are possible. Double-check key information.

Question: How did my plant perform yesterday?

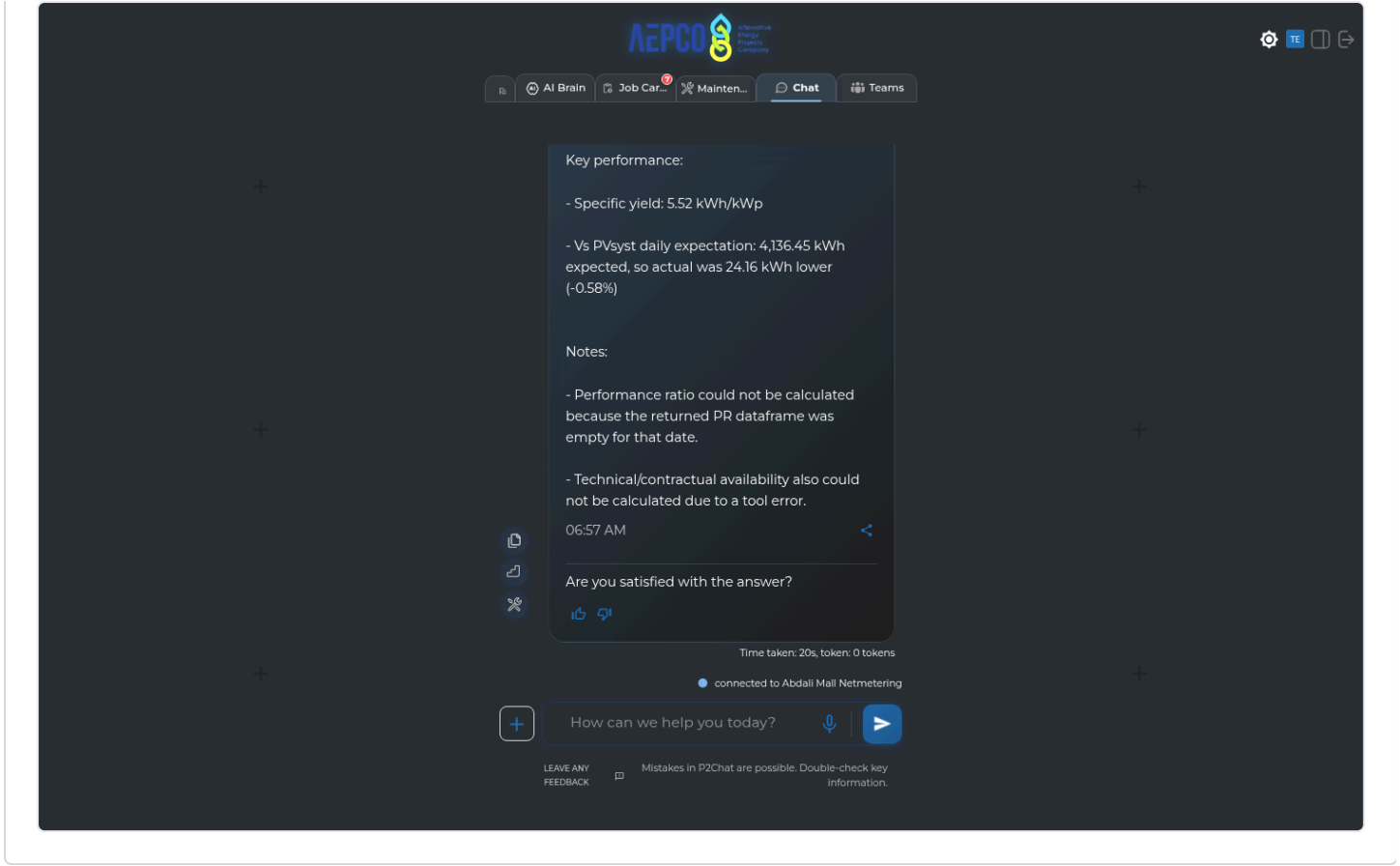
Error reason: AI indicated failure (matched 'tool error'): Yesterday (2026-08-09), your plant produced 4,112.29 kWh. Key performance: - Specific yield: 5.52 kWh/kWp - Vs PVsyst daily expectation: 4,136.45 kWh expected, so actual was 24.16 kWh lower (-0.58%)

TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "Yesterday (2026-08-09), your plant produced 4,112.29 kWh.\n\nKey performance:\n- Specific yield: 5.52 kWh/kWp\n- Vs PVsyst daily expectation: 4,136.45 kWh expected, so actual was 24.16 kWh lower (-0.58%)\n\nNotes:\n- Performance ratio could not be calculated because the returned PR dataframe was empty for that date.\n- Technical/contractual availability also could not be calculated due to a tool error.",
    "coding_response": [
      "Yesterday (2026-08-09), your plant produced 4,112.29 kWh.\n\nKey performance:\n- Specific yield: 5.52 kWh/kWp\n- Vs PVsyst daily expectation: 4,136.45 kWh expected, so actual was 24.16 kWh lower (-0.58%)\n\nNotes:\n- Performance ratio could not be calculated because the returned PR dataframe was empty for that date.\n- Technical/contractual availability also could not be calculated due to a tool error."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "functions.pvsyst_expectation",
      "multi_tool_use.parallel"
    ],
    "urls": [],
    "steps": "1. Select data from 2026-08-09 to 2026-08-09, including the end date.\n2. Calculate the total energy produced by the plant for each month.\n3. Check that the monthly totals are correct and that each month ends on the last day.\n4. Get the plant's theoretical P50 energy targets for each month from the metadata.\n5. Combine the actual monthly energy data with the P50 targets using the month names.\n6. For each month, calculate performance as a percentage: (actual energy / P50 energy) * 100.\n7. Add up the actual and P50 energy values over the whole period.\n8. Calculate overall performance for the period as a percentage: (total actual energy / total P50 energy) * 100.\n9. Make sure all months are included in the results.\n10. Include all calculated columns in the final output.\n11. Return the results as a dictionary, using the index for easy lookup.\n\n---\n\n1. Select data from 2026-08-09 to 2026-08-09, including the end date.\n2. Group the data by day (or by minute, hour, month, or year as needed).\n3. Make sure every day in the range is included.\n4. Calculate the total energy produced during the period using the plant's power data, adjusting for a 0.0833-hour interval.\n5. Divide the total energy by the plant's rated capacity to get the specific yield in kWh per kWp.\n6. Return the specific yield for each time interval as a detailed breakdown.\n\n---\n\n1. Select data from 2026-08-09 to 2026-08-09, including the end date.\n2. Use the active_power column(s) for calculations.\n3. Clean the data by removing or marking invalid power readings.\n4. For each data point, calculate energy as (Power * 0.0833 hours) and convert units if needed (e.g., from Watts to kW).\n5. Include all days in the calculation.\n6. Add up all energy values in the period to get the total energy produced.\n\n---\n\n1. Select data from 2026-08-09 to 2026-08-09, including the end date.\n2. Find all inverter power columns and their DC rated power from the plant's metadata.\n3. Set calculation parameters: irradiance threshold (usually 50 W/m²), power threshold (usually 0.1 kW), and data interval (typically 5 minutes).\n4. For each inverter, filter data to times when irradiance is above the threshold, then count the number of useful and downtime intervals.\n5. Convert these counts to hours based on the data interval.\n6. Calculate technical availability for each inverter: ((useful time - downtime) / useful time) * 100.\n7. Return a dictionary with the availability percentage for each inverter.\n\n---\n\n1. Select data from 2026-08-09 to 2026-08-09, including the end date.\n2. Identify the columns for power, irradiance, and temperature.\n3. Clean the data by removing or flagging invalid power and irradiance values.\n4. Calculate the actual energy produced by multiplying power by the time interval (0.0833 hours) and adjusting units as needed.\n5. Calculate
```



Graph/Visualization Errors GRAPH VISUALIZATION (1 failure)

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Error reason: AI indicated failure (matched 'unable to provide'): - Clipping loss breakdown by inverter (last 6 months): • Unable to provide data due to an internal calculation error (unsupported operand type(s) for /: 'NoneType' and 'int'). - Visualization:

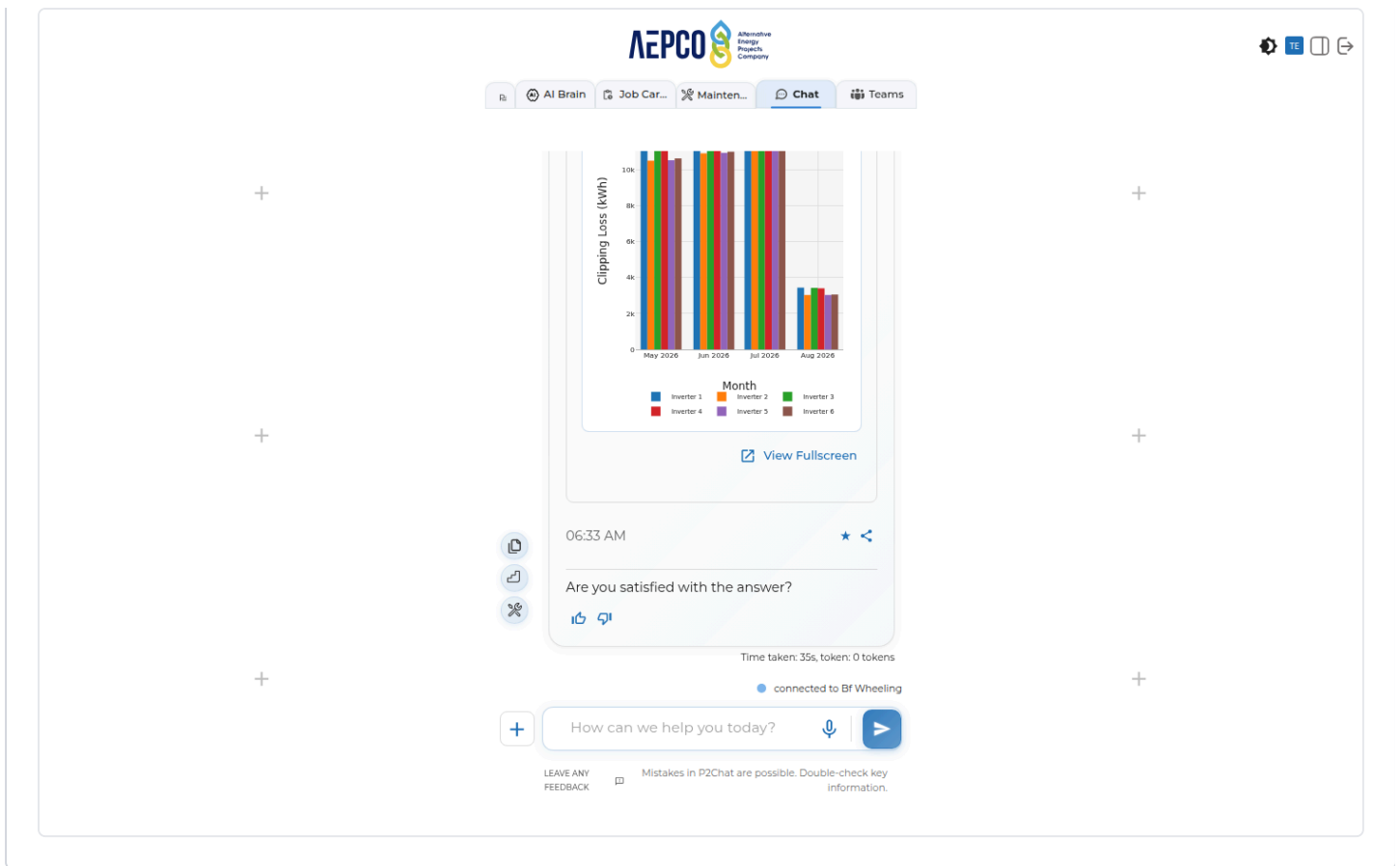
TECHNICAL EVIDENCE

API call: GET /testuser2/ai_agent_chat → 200

Raw API response:

```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well",
    "displayed_response": "- Clipping loss breakdown by inverter (last 6 months):\n • Unable to provide data due to an internal calculation error (unsupported operand type(s) for /: 'NoneType' and 'int').\n\n- Visualization:\n • The chart for monthly clipping loss by inverter (last 6 months) has been prepared.",
    "coding_response": [
      "I couldn't calculate the clipping loss because the clipping-loss tool failed internally for this 6-month range with this error: `unsupported operand type(s) for /: 'NoneType' and 'int'` while building the digital twin. I'm not able to provide a trustworthy inverter breakdown from that failed run.",
      {
        "type": "graph",
        "series": [
          "<html>\n<head><meta charset='utf-8' /></head>\n<body>\n  <div>
            <script type='text/javascript'>window.PlotlyConfig = {MathJaxConfig: 'local'};</script>\n      <script charset='utf-8' src='https://cdn.plot.ly/plotly-3.0.1.min.js'></script>
            <div id='b3e32d6c-db8c-41bf-8dd7-62fe4d11299b' style='height:100%; width:100%;'></div>
            <script type='text/javascript'>
              window.PLOTLYENV=window.PLOTLYENV || {};
              if (document.getElementById('b3e32d6c-db8c-41bf-8dd7-62fe4d11299b')) {
                Plotly.newPlot(
                  'b3e32d6c-db8c-41bf-8dd7-62fe4d11299b',
                  [{
                    'x': 'Inverter 1',
                    'y': 2.2,
                    'type': 'bar',
                    'hovertemplate': '%{fullData.name}: %y: 2.2 kWh'
                  }, {
                    'x': 'Inverter 2',
                    'y': 2.2,
                    'type': 'bar',
                    'hovertemplate': '%{fullData.name}: %y: 2.2 kWh'
                  }, {
                    'x': 'Inverter 3',
                    'y': 2.2,
                    'type': 'bar',
                    'hovertemplate': '%{fullData.name}: %y: 2.2 kWh'
                  }, {
                    'x': 'Inverter 4',
                    'y': 2.2,
                    'type': 'bar',
                    'hovertemplate': '%{fullData.name}: %y: 2.2 kWh'
                  }, {
                    'x': 'Inverter 5',
                    'y': 2.2,
                    'type': 'bar',
                    'hovertemplate': '%{fullData.name}: %y: 2.2 kWh'
                  }, {
                    'x': 'Inverter 6',
                    'y': 2.2,
                    'type': 'bar',
                    'hovertemplate': '%{fullData.name}: %y: 2.2 kWh'
                  }
                ],
                {
                  'barmode': 'group',
                  'hovermode': 'x'
                },
                {
                  'orientation': 'h',
                  'xanchor': 'center',
                  'yanchor': 'top',
                  'font': {
                    'size': 8
                  },
                  'itemwidth': 30,
                  'bgcolor': 'rgba(0,0,0,0)',
                  'bordercolor': 'rgba(0,0,0,0)'
                },
                {
                  'data': {
                    'scatter': [
                      {
                        'type': 'scatter'
                      }
                    ]
                  },
                  'title': 'Monthly clipping loss by inverter (last 6 mo'
                }
              ]
            }
          ]
        }
      ]
    }
  }
}
```

Console errors: An iframe which has both allow-scripts and allow-same-origin for its sandbox attribute can escape its sandboxing.



Skipped (42)

Plant	Question	Env	Skip reason
SADAFCO	Please generate last months report	Development	Plant connection failed
SADAFCO	Based on pvsyst report, what is the rated power of this plant and array configuration	Development	Plant connection failed
SADAFCO	Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2	Development	Plant connection failed
SADAFCO	Please plot last 3 days hourly performance ratio for my Strings for inverter 2	Development	Plant connection failed
SADAFCO	Please plot the last 4 days hourly performance ratio for all my inverters	Development	Plant connection failed
SADAFCO	Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3	Development	Plant connection failed
SADAFCO	Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)	Development	Plant connection failed
SADAFCO	How can I shut down my inverters?	Development	Plant connection failed

Plant	Question	Env	Skip reason
SADAFCO	I have a COMMUNICATION_FAULT what do I do to my inverters?	Development	Plant connection failed
SADAFCO	When was the last time the BOM data showed up	Development	Plant connection failed
SADAFCO	Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well	Development	Plant connection failed
SADAFCO	Was there any 0 production days in the last 7 days?	Development	Plant connection failed
SADAFCO	Please change to dark mode	Development	Plant connection failed
SADAFCO	Pull up the latest cleaning log	Development	Plant connection failed
SADAFCO	Jim cleaned the pyranometer today, can you add it to the cleaning log?	Development	Plant connection failed
SADAFCO	Please generate December 2025 performance report	Development	Plant connection failed
SADAFCO	How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?	Development	Plant connection failed
SADAFCO	How did my plant perform yesterday?	Development	Plant connection failed
SADAFCO	Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt	Development	Plant connection failed
SADAFCO	Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times	Development	Plant connection failed
SADAFCO	Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well	Development	Plant connection failed
SAK_1	Please generate last months report	Development	Plant connection failed
SAK_1	Based on pvsyst report, what is the rated power of this plant and array configuration	Development	Plant connection failed
SAK_1	Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2	Development	Plant connection failed
SAK_1	Please plot last 3 days hourly performance ratio for my Strings for inverter 2	Development	Plant connection failed

Plant	Question	Env	Skip reason
SAK_1	Please plot the last 4 days hourly performance ratio for all my inverters	Development	Plant connection failed
SAK_1	Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3	Development	Plant connection failed
SAK_1	Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)	Development	Plant connection failed
SAK_1	How can I shut down my inverters?	Development	Plant connection failed
SAK_1	I have a COMMUNICATION_FAULT what do I do to my inverters?	Development	Plant connection failed
SAK_1	When was the last time the BOM data showed up	Development	Plant connection failed
SAK_1	Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well	Development	Plant connection failed
SAK_1	Was there any 0 production days in the last 7 days?	Development	Plant connection failed
SAK_1	Please change to dark mode	Development	Plant connection failed
SAK_1	Pull up the latest cleaning log	Development	Plant connection failed
SAK_1	Jim cleaned the pyranometer today, can you add it to the cleaning log?	Development	Plant connection failed
SAK_1	Please generate December 2025 performance report	Development	Plant connection failed
SAK_1	How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?	Development	Plant connection failed
SAK_1	How did my plant perform yesterday?	Development	Plant connection failed
SAK_1	Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt	Development	Plant connection failed
SAK_1	Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times	Development	Plant connection failed
SAK_1	Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well	Development	Plant connection failed

